



New-Generation Pesticides and Their Ecological Consequences in Aquatic Environments

A Yamini¹, R Arasi² and P Anushiya³

Department of Aquatic Environment Management

Fisheries College and Research Institute, Thoothukudi,

Tamilnadu Dr.J.Jayalalithaa Fisheries University,

Tamilnadu- 628 008.

Abstract

New-generation pesticides, including neonicotinoids, pyrethroids, and systemic herbicides, have been widely adopted for their improved efficacy and targeted action. However, their presence in aquatic environments raises significant ecological concerns. These pesticides can enter water bodies through agricultural runoff, wastewater discharge, and aquaculture practices, leading to unintended toxic effects on non-target organisms. Their persistence, bioaccumulation, and biomagnification further exacerbate risks to aquatic biodiversity, disrupting food webs and microbial communities. Moreover, antibiotic contamination fosters antimicrobial resistance, posing additional threats to ecosystem health. Sub-lethal exposures can impair behaviour, physiology, and immune responses in aquatic species. To mitigate these adverse impacts, regulatory measures, enhanced monitoring, and alternative pest management strategies, such as integrated pest management and bioremediation, are essential. Balancing agricultural productivity with environmental sustainability requires a comprehensive approach to pesticide use and ecological conservation.

Introduction:

New-generation pesticides have been lauded for their improved efficacy, targeted application, and reduced environmental persistence compared to traditional pesticides. However, these advancements do not eliminate their ecological consequences, particularly in aquatic ecosystems. Aquatic environments are highly sensitive to chemical contaminants due to the interconnectedness of aquatic food webs, the dependency of various species on water quality, and the cumulative effects of pollutants. The increasing use of new-generation antibiotics to combat resistant pathogens has raised concerns about their environmental impact, particularly in aquatic ecosystems. Antibiotics enter water bodies through various pathways, including wastewater discharge, agricultural runoff, and aquaculture practices, where they may exert toxic effects on non-target organisms.



Types of New Generation Pesticides

New generation pesticides are derived from naturally occurring substances such as bacteria, fungi, and plant extracts. They include microbial pesticides, which utilize beneficial microbes to combat harmful pests without negatively impacting other organisms. RNA interference (RNAi)-based pesticides work by silencing specific genes in pests, preventing them from reproducing or feeding effectively. Nanopesticides use nanotechnology to enhance the controlled release and absorption of active ingredients, minimizing chemical runoff and environmental pollution. Botanical pesticides, extracted from plants with natural insecticidal properties such as neem oil and pyrethrin, are increasingly used as sustainable alternatives in organic farming. These advanced pesticides offer more targeted approaches to pest control while reducing the negative impact on ecosystems and human health.

Effects on the Aquatic Ecosystem

While new generation pesticides are designed to be more environmentally friendly, their impact on aquatic ecosystems must be carefully assessed. Runoff from agricultural fields can carry these pesticides into water bodies, potentially affecting aquatic organisms. Some nanopesticides, despite their controlled release mechanisms, may still accumulate in aquatic environments and alter the physiological functions of fish and invertebrates. The nano-sized particles can pass through biological membranes more easily, leading to bioaccumulation and toxicity in aquatic food chains. Studies suggest that prolonged exposure to nanopesticides can disrupt reproductive and metabolic processes in fish, leading to long-term ecological imbalances.

RNAi-based pesticides, though highly specific, might unintentionally interfere with non-target aquatic organisms if their genetic sequences are similar to those of targeted pests. This can impact essential aquatic species such as plankton, which serve as the base of the food web, potentially leading to cascading effects throughout the ecosystem. Moreover, the degradation process of these RNAi molecules in water bodies is not yet fully understood, posing concerns about their persistence and unforeseen genetic disruptions in aquatic life.

Microbial pesticides, while generally safer, could disrupt the balance of aquatic microbial communities. Since these pesticides introduce foreign bacteria or fungi to control pests, there is a risk that they might interact with native aquatic microbes, altering nutrient cycles and potentially leading to harmful algal blooms. Changes in microbial populations can affect oxygen levels in water bodies, leading to hypoxic conditions that threaten fish and other marine organisms. Even botanical pesticides, which are often considered the safest alternative, can be problematic if overused. Some plant-derived compounds can be toxic to fish and amphibians, especially in enclosed or slow-moving water bodies where dilution is limited. For instance, high concentrations of neem-based pesticides have been reported to cause physiological stress in freshwater fish, affecting their immune response and growth rates.



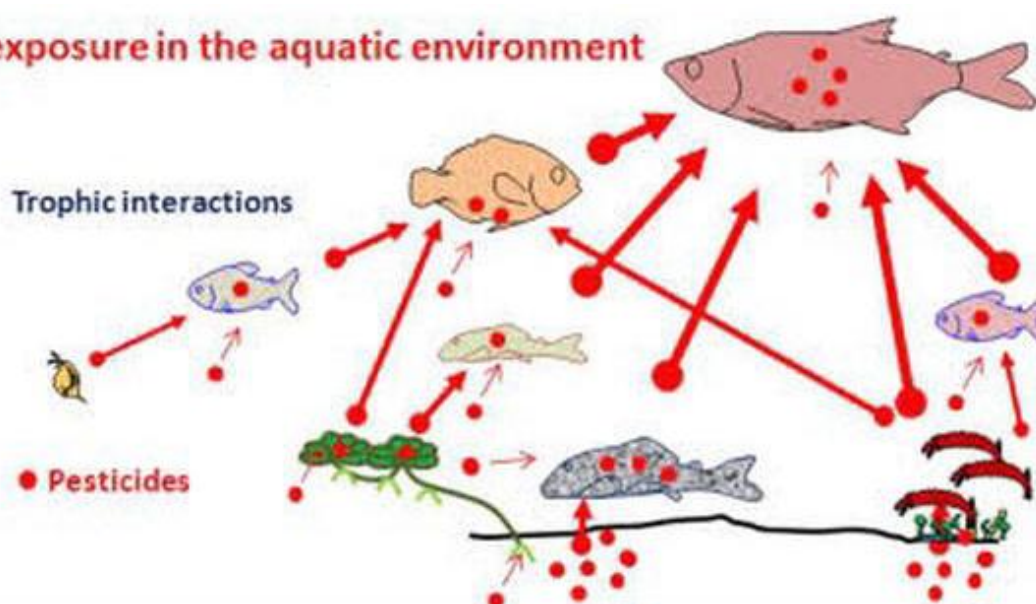


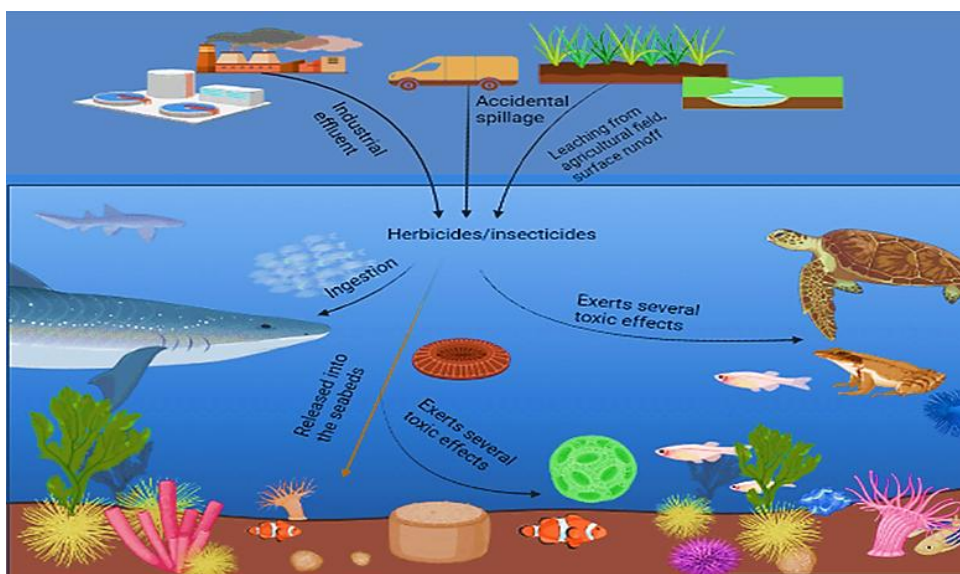
Bioaccumulation and Biomagnification

New-generation pesticides, especially those with high chemical stability like neonicotinoids, pyrethroids, and certain organophosphates, pose a growing risk of bioaccumulation and biomagnification in aquatic ecosystems. Bioaccumulation refers to the gradual build-up of these toxic compounds in the tissues of aquatic organisms over time, especially in species like fish, mollusks, and crustaceans that are directly or indirectly exposed through contaminated water or sediment. Many of these compounds are lipophilic, meaning they dissolve in fat and tend to stay in the body rather than being excreted, leading to long-term exposure even after pesticide levels in the water drop. Biomagnification, on the other hand, is when these accumulated toxins amplify through the food chain—from plankton to small fish, then to larger predators like birds or humans who consume aquatic species.

In India, studies from river basins like the Ganga and Yamuna have shown traces of persistent pesticide residues in fish and sediments, hinting at this silent toxic buildup. Even small concentrations at the base of the food web can multiply many times by the time they reach top predators, affecting reproduction, growth, and survival of species. This not only disrupts aquatic biodiversity but also threatens food safety and human health. With increasing use of systemic and long-acting pesticides in agriculture, there's an urgent need for continuous monitoring and adoption of safer alternatives to break this toxic chain before it spirals out of control.

Fish exposure in the aquatic environment





Reducing Pesticide Resistance

Another crucial benefit of new generation pesticides is their role in reducing pesticide resistance among pests. Traditional pesticides often lead to resistance as pests evolve to withstand chemical treatments, necessitating higher doses or stronger formulations over time. This has led to an increase in pesticide use, further exacerbating environmental and health risks. However, new generation pesticides, with their novel modes of action, significantly reduce the likelihood of resistance development. RNAi-based pesticides work by interfering with essential genes in pests, making it extremely difficult for them to develop resistance since they target specific genetic sequences. This specificity ensures that the pesticide remains effective for longer periods without requiring increased dosages. Additionally, since RNAi molecules degrade quickly in the environment, they do not persist long enough to allow pests to adapt. Microbial pesticides, derived from naturally occurring bacteria, fungi, or viruses, introduce biological control agents that pests find challenging to resist. These biological pesticides work through multiple mechanisms, such as infecting pests, producing toxins, or disrupting their digestive systems. Because microbial pesticides involve living organisms, pests are less likely to develop widespread resistance compared to chemical pesticides.

Botanical pesticides contain complex mixtures of bioactive compounds, making it difficult for pests to develop resistance. Unlike synthetic pesticides that rely on single active ingredients, botanical pesticides work through multiple biochemical pathways, ensuring long-term pest control. Neem-based pesticides, for example, contain azadirachtin, which affects insect growth and reproduction in several ways, making resistance highly unlikely.

Role in Integrated Pest Management (IPM)

Another critical aspect of new generation pesticides is their potential role in integrated pest management (IPM) systems. IPM is a holistic approach to pest control that combines biological, cultural, mechanical, and chemical methods to minimize pesticide use while



ensuring effective pest control. New generation pesticides fit well into this approach by providing safer and more sustainable alternatives to traditional chemical treatments. By integrating these pesticides with other IPM strategies, such as crop rotation, habitat manipulation, and the use of natural predators, farmers can achieve long-term pest control while minimizing environmental impact. This not only enhances agricultural sustainability but also contributes to global efforts to reduce the negative effects of conventional pesticides on human health and the environment.

Mitigation Strategies and Future Directions

New-generation pesticides, while effective against pests, are causing serious ecological stress in aquatic environments due to their high toxicity and persistence. To tackle this, several mitigation strategies are being explored. One of the most practical is the use of buffer zones and vegetative barriers around agricultural fields, which help filter out pesticides before they enter nearby water bodies. Another eco-friendly method is bioremediation, where microbes or aquatic plants like duckweed are used to break down harmful chemicals in contaminated water. Precision pesticide application using drones and AI tools is also gaining traction, ensuring minimal usage and reducing runoff. Integrated Pest Management (IPM), which combines biological, mechanical, and minimal chemical approaches, promotes sustainable pest control with reduced environmental impact. Constructed wetlands are another powerful solution—they act as natural purifiers, filtering pesticide-laden runoff through engineered ecosystems.

Looking ahead, the future lies in developing biodegradable and eco-safe pesticide formulations, such as nano-pesticides and plant-based bio-pesticides. Strengthening policies and regulations around pesticide usage and water monitoring is equally important, especially in India where data from local water bodies is still patchy. Emerging technologies like AI and big data can help track water quality in real-time through smart sensors. Public awareness also plays a huge role—training farmers, revamping curriculum, and even using social media can create a much-needed cultural shift. Ultimately, a collaborative approach involving scientists, farmers, policymakers, and tech innovators is key. On the cost front, solutions like bioremediation setups range from ₹50,000 to ₹2 lakh, drone-based spraying is around ₹400–₹700 per acre, while constructed wetlands can go up to ₹5 lakh. It's high time we move from damage control to damage prevention and rethink how we protect our water while feeding the nation.

Conclusion:

New-generation pesticides, though designed for better efficiency and targeted pest control, pose a significant risk to aquatic ecosystems due to their chemical complexity, persistence, and bioaccumulation potential. Their unchecked use leads to disruption of aquatic food chains, toxicity in non-target organisms, and degradation of water quality, affecting both biodiversity and human health. While technological advancements and agricultural needs justify their development, it's crucial to strike a balance between productivity and sustainability.





Moving forward, adopting eco-friendly farming practices, enforcing strict regulations, promoting research into biodegradable alternatives, and enhancing awareness among stakeholders are essential. Only through an integrated, multidisciplinary approach can we minimize the environmental footprint of these pesticides and ensure the long-term health of our aquatic ecosystems in India and beyond.

References:

1. Aktar, M. W., Sengupta, D., & Chowdhury, A. (2009). Impact of pesticide use in agriculture: Their benefits and hazards. *Interdisciplinary Toxicology*, 2(1), 1–12. <https://doi.org/10.2478/v10102-009-0001-7>
2. Chen, J., Han, F., Xu, T., & Gao, L. (2023). Antibiotic resistance in aquaculture and aquatic organisms: A review. *Frontiers in Microbiology*, 14(1).
3. Chowdhury, A., Pradhan, S., Saha, M., & Sanyal, N. (2019). Pesticide residues in water – A global scenario with special reference to India. *Environmental Monitoring and Assessment*, 191(10), 575. <https://doi.org/10.1007/s10661-019-7742-7>
4. Kumar, V., & Sharma, A. (2021). Antibiotics: An overview on the environmental occurrence, toxicity, and microbial resistance. *Critical Reviews in Environmental Science and Technology*, 51(9), 869-908. □
5. Liu, Y., Wang, H., & Qian, Y. (2023). Antimicrobial transformation products in the aquatic environment. *Environmental Science & Technology*, 57(5), 3123-3140.
6. Malaj, E., von der Ohe, P. C., Grote, M., Kühne, R., Mondy, C. P., Usseglio-Polatera, P., ... & Schäfer, R. B. (2014). Organic pesticides threaten aquatic biodiversity in Europe and the United States. *Proceedings of the National Academy of Sciences*, 111(28), 9549–9554. <https://doi.org/10.1073/pnas.1321082111>
7. Rico, A., Satapornvanit, K., Haque, M. M., Min, J., Nguyen, P. T., Telfer, T. C., & Van den Brink, P. J. (2010). Use of species sensitivity distributions to assess the ecological risk of pesticides to aquatic life in Southeast Asia. *Ecotoxicology and Environmental Safety*, 73(7), 1040–1046.
8. Xia, W., Liu, Z., & He, J. (2023). Impact of antibiotic pollution on the bacterial population within aquatic environments. *Water*, 15(5), 975.
9. Yang, L., Zhang, Y., Guo, X., & Xu, Y. (2023). Ecotoxicological effects of microplastics combined with antibiotics. *Environmental Toxicology and Chemistry*, 42(2), 495-512.
10. Zhang, Q., Feng, Y., & Ng, T. K. (2017). Editorial: Antibiotic resistance in aquatic systems. *Frontiers in Microbiology*, 8(1), 14.

